

Independence

Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$ $P(B) > 0$

Def 1: 2 events, A, B are independent if $P(A \cap B) = P(A) \cdot P(B)$

Def 2: 2 events, A, B are independent if $P(A|B) = P(A)$

Def 1 $\Leftrightarrow$ Def 2

Def 1 $\rightarrow$ Def 2: $P(A \cap B) = P(A) \cdot P(B)$

By multiplication rule $P(A \cap B) = P(A|B) \cdot P(B)$

then $P(A) \cdot P(B) = P(A|B) \cdot P(B)$

then $P(A) = P(A|B)$

Def 2 $\rightarrow$ Def 1: $P(A|B) = P(A)$

By multiplication rule $P(A \cap B) = P(A|B) \cdot P(B)$

By def 1 $P(A|B) \cdot P(B) = P(A \cap B) = P(A) \cdot P(B)$

EX 1 /

①	②
③	④

 Take 1 chip $S = \{1, 2, 3, 4\}$
Event: $A = \{1, 2\}$ $B = \{1, 3\}$
 $P(A) = 1/2$ $P(B) = 1/2$

Are events independent? $P(A \cap B) = P(A) \cdot P(B)$

$P(A \cap B) = 1/4$ $P(A) \cdot P(B) = 1/4$ These events are independent

EX 2 /

①	②
③	④

 Take 1 chip $S = \{1, 2, 3, 4, 5\}$
 $A = \{1, 2\}$ $B = \{1, 3\}$ $P(A) = 2/5$ $P(B) = 2/5$

$A \cap B = \{1\}$ $P(A \cap B) = 1/5$ Does $P(A \cap B) = P(A) \cdot P(B)$?

These events are not independent

EX 3 Let A and B be indep. What can be said about A^c and B^c

Prop: If A and B are indep. then A^c and B^c are indep.

Assume $P(A \cap B) = P(A) \cdot P(B)$

$$P((1 - P(A))(1 - P(B))) = P(A^c) \cdot P(B^c)$$

$$1 - (P(A) + P(B) - P(A) \cdot P(B)) = 1 - P(A) - P(B) + P(A) \cdot P(B) = P(A^c) \cdot P(B^c)$$

$$\text{Then } P(A \cup B)^c = 1 - P(A \cup B) = 1 - (P(A) + P(B) - P(A \cap B))$$

Then $A^c \cap B^c = (A \cup B)^c$ by demorgans

$$\text{Then } P(A^c \cap B^c) = P(A^c) \cdot P(B^c)$$